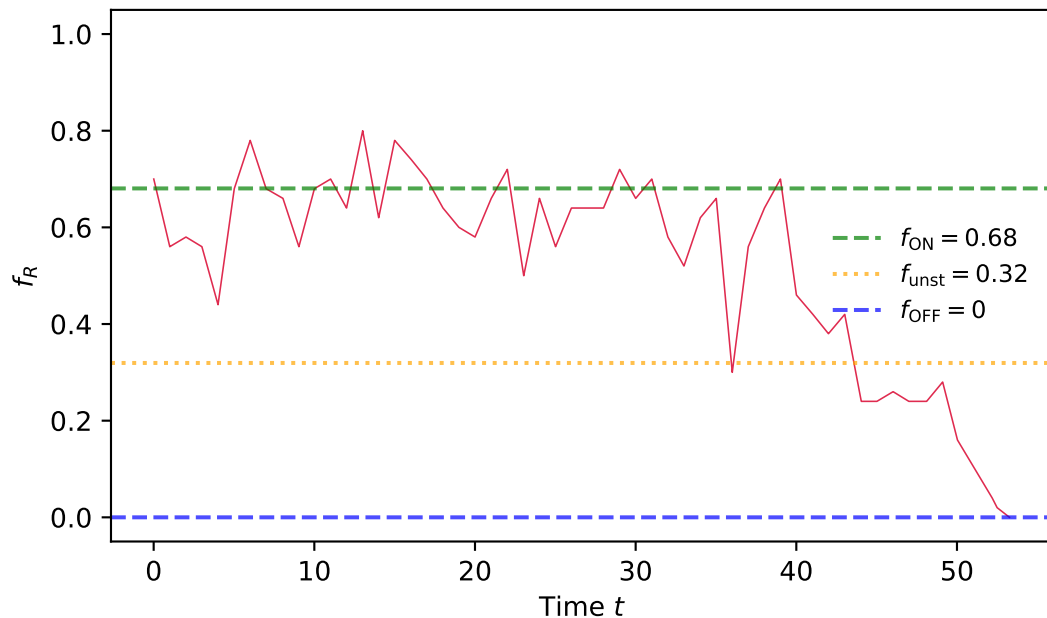
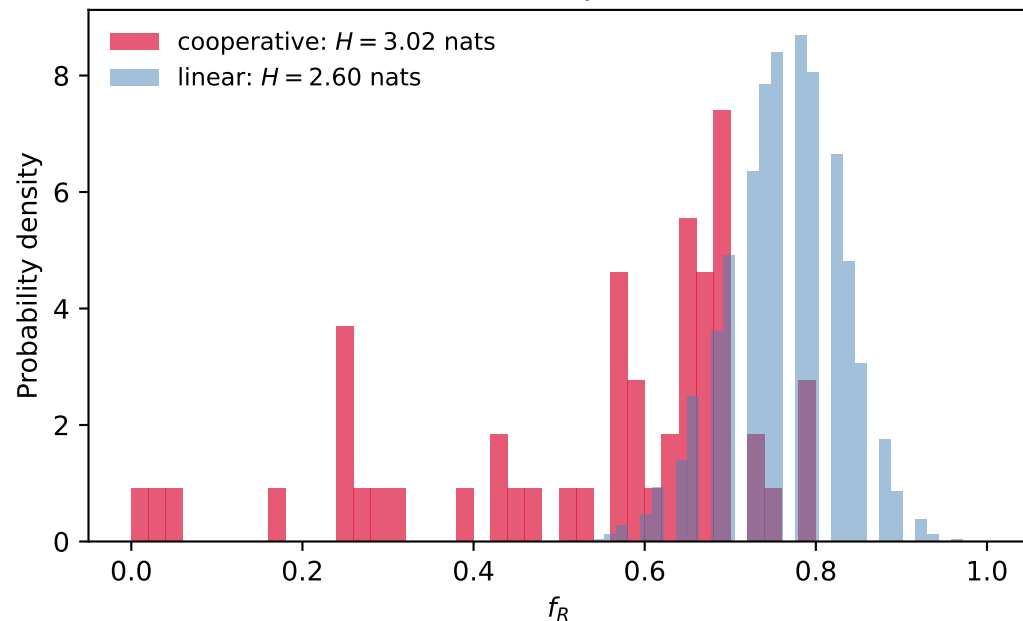
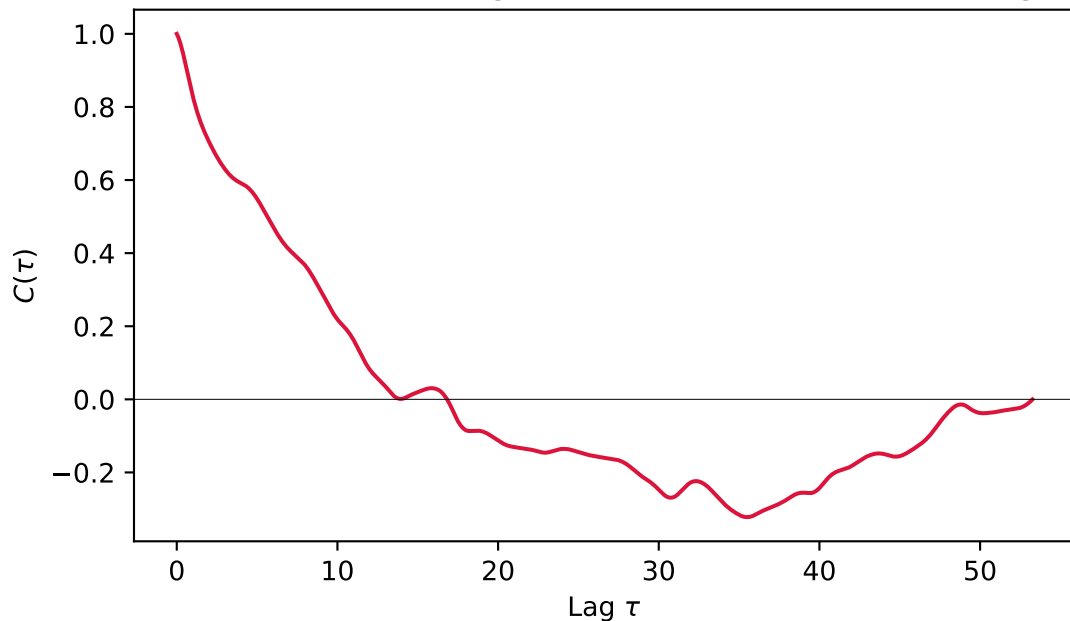


Cooperative ant (bistable, $g = 4.6$): rare switches

State distribution: cooperative is bimodal



Autocorrelation: long-time tail from bistable switching

**State-space entropy reduction**

Cooperative ant, $V = 50$, $g = 4.6$:

$$H_{\text{uniform}} = \ln(V + 1) = 3.93 \text{ nats}$$

$$H(f_R)_{\text{cooperative}} = 3.02 \text{ nats}$$

$$\Delta H = 0.92 \text{ nats} (= 23\% \text{ of uniform})$$

$$H(f_R)_{\text{linear}} = 2.60 \text{ nats}$$

The canonical ant does NOT form spatial trails (no biased movement, by design). Instead: bistable state-space. The cooperative variant sits in ON or OFF for long times; rare switches between basins are governed by the CFAC instanton action $S^* = 0.0265$.